Proxmox Ceph and ZFS Implementation Report

Three-node cluster: prox22, prox23 and prox24

Document type	Implementation record and operational SOP
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Cluster	cisai-pve
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Scope	Ceph rebuild, network repair, RAIDZ2 deployment, Proxmox storage integration and physical disk mapping
Final status	HEALTHY / OPERATIONAL

Final implemented design: 12 Ceph OSDs for shared RBD storage, plus one independent seven-disk RAIDZ2 pool on each Proxmox node.

1. Executive summary

The existing Ceph disk configuration was removed and rebuilt across the three-node Proxmox cluster. Four 20 TB SAS disks per node were assigned to Ceph, producing 12 OSDs. Seven additional 20 TB SAS disks per node were assigned to independent local ZFS RAIDZ2 pools. The remaining 20 TB disk on each node was retained as reserved capacity.

Component	Implemented state
Proxmox cluster	cisai-pve, 3 nodes, quorate
Ceph network	10.31.122.0/24 on nic2
Ceph MON	prox22, prox23 and prox24 in quorum
Ceph OSD	12 up, 12 in, four per node
Ceph RBD pool	ceph, size 3, min_size 2
Ceph PG state	33 active+clean
Ceph health	HEALTH_OK
ZFS	Three independent seven-disk RAIDZ2 pools
ZFS state	ONLINE, zero read/write/checksum errors
PERC exposure	12 large HDDs presented as JBOD; OS SSD pair in RAID1

Important: The three ZFS pools are independent node-local storage. They are not shared storage. HA and shared VM disks should remain on the Ceph RBD storage.

2. Final architecture

Node	Management/cluster role	Ceph interface	Ceph OSDs	Local ZFS pool
prox22	Proxmox node, MON, standby MGR	10.31.122.22/24	osd.0, osd.3, osd.6, osd.9	local-zfs-prox22/pve
prox23	Proxmox node, MON, active MGR	10.31.122.23/24	osd.1, osd.4, osd.7, osd.10	local-zfs-prox23/pve
prox24	Proxmox node, MON leader	10.31.122.24/24	osd.2, osd.5, osd.8, osd.11	local-zfs-prox24/pve

Disk role per node

Device range	Role
/dev/sda - /dev/sdd	Ceph BlueStore OSDs
/dev/sde - /dev/sdk	Seven-disk local ZFS RAIDZ2
/dev/sdl	Reserved / not in a pool
/dev/sdm	Proxmox OS virtual disk backed by two SSDs in PERC RAID1

3. Initial safety checks and inventory

The OS disk and storage disks were identified before destructive operations. On every node, root was /dev/mapper/pve-root, EFI was /dev/sdm2, and the OS disk was /dev/sdm.

```
hostname
findmnt -no SOURCE /
findmnt -no SOURCE /boot/efi
```

```
lsblk -o NAME,PATH,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL,WWN
pvs
vgs
lvs
ceph-volume lvm list
```

Timestamped evidence backup

```
BACKUP_DIR="/root/ceph-before-zap-$(hostname -s)-$(date +%F-%H%M%S)"
mkdir -p "$BACKUP_DIR"
hostname > "$BACKUP_DIR/hostname.txt"
ceph-volume lvm list > "$BACKUP_DIR/ceph-volume-lvm-list.txt"
lsblk -o NAME,PATH,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL,WWN > "$BACKUP_DIR/lsblk.txt"
pvs -o+pv_uuid > "$BACKUP_DIR/pvs.txt"
vgs -o+vg_uuid > "$BACKUP_DIR/vgs.txt"
lvs -a -o+devices,lv_tags > "$BACKUP_DIR/lvs.txt"
ceph -s > "$BACKUP_DIR/ceph-status.txt" 2>&1 || true
ceph osd tree > "$BACKUP_DIR/ceph-osd-tree.txt" 2>&1 || true
```

4. Old Ceph OSD removal and disk cleanup

Detect and stop local OSD services

```
mapfile -t OSD_IDS < <(
  ceph-volume lvm list | sed -n 's/^====== osd\.\([0-9]\+\).*$/■/p'
)
printf 'osd.%s
' "${OSD_IDS[@]}"
for id in "${OSD_IDS[@]}"; do
  systemctl stop "ceph-osd@${id}.service"
  systemctl disable "ceph-osd@${id}.service" 2>/dev/null || true
done
pgrep -a ceph-osd || echo "No ceph-osd processes are running"
```

Zap the former Ceph disks

```
TARGET_DISKS=(
 /dev/sda /dev/sdb /dev/sdc /dev/sdd
 /dev/sde /dev/sdf /dev/sdg /dev/sdh
 /dev/sdi /dev/sdj /dev/sdk /dev/sdl
)
for disk in "${TARGET_DISKS[@]}"; do
  ceph-volume lvm zap "$disk" --destroy
done
udevadm settle
partprobe || true
pvscan --cache
```

Safety control: /dev/sdm was never included. Glob patterns such as /dev/sd? were avoided because they would also match the OS disk.

Post-cleanup validation

```
pvs
vgs
lvs
lsblk -o NAME,PATH,SIZE,TYPE,FSTYPE,MOUNTPOINTS /dev/sd{a..m}
for disk in /dev/sd{a..l}; do wipefs --no-act "$disk"; done
ceph-volume lvm list
pgrep -a ceph-osd || echo "No ceph-osd processes are running"
```

5. prox22 Ceph monitor network repair

mon.prox22 failed because Ceph was configured to bind to 10.31.122.22, but nic2 had no IPv4 address. The persistent network configuration was corrected while the management network on vmbr0 remained unchanged.

Backup and persistent configuration

```
BACKUP_DIR="/root/network-backup-$(date +%F-%H%M%S)"
mkdir -p "$BACKUP_DIR"
cp -a /etc/network/interfaces "$BACKUP_DIR/interfaces"
cp -a /etc/pve/ceph.conf "$BACKUP_DIR/ceph.conf" 2>/dev/null || true

# /etc/network/interfaces
auto nic2
iface nic2 inet static
    address 10.31.122.22/24
```

Apply and verify

```
ifup nic2
ip -br address show nic2
ip route show dev nic2
ip route get 10.31.122.23
ip route get 10.31.122.24
systemctl reset-failed ceph-mon@prox22.service
systemctl enable ceph-mon@prox22.service
systemctl restart ceph-mon@prox22.service
systemctl is-active ceph-mon@prox22.service
ceph mon stat
ceph -s
```

The monitor rejoined quorum successfully: prox24, prox22 and prox23.

6. Obsolete OSD record cleanup and 12-OSD creation

Old OSD map entries were marked out/down, stopped, and purged. A newly created temporary osd.36 was also removed so the clean allocation could restart. Ceph then allocated IDs cluster-wide as the new disks were created.

```
ceph osd out <OSD-ID>
ceph osd down <OSD-ID>
systemctl stop ceph-osd@<OSD-ID>.service
ceph osd purge <OSD-ID> --yes-i-really-mean-it
```

Create four OSDs per node

```
# Run locally on prox22, then prox23, then prox24
pveceph osd create /dev/sda
pveceph osd create /dev/sdb
pveceph osd create /dev/sdc
pveceph osd create /dev/sdd

ceph-volume lvm list
ceph osd tree
ceph osd stat
ceph -s
```

Node	Disk	OSD ID
prox22	/dev/sda, /dev/sdb, /dev/sdc, /dev/sdd	0, 3, 6, 9
prox23	/dev/sda, /dev/sdb, /dev/sdc, /dev/sdd	1, 4, 7, 10
prox24	/dev/sda, /dev/sdb, /dev/sdc, /dev/sdd	2, 5, 8, 11

7. Stale PG diagnosis and pool recreation

After OSD creation, all 12 OSDs were up and in, but 33 persistent PGs were unknown. CRUSH mapping was valid, while blocked operations remained queued for PG. The stale pools were removed and recreated because the deployment was an intentional fresh rebuild.

Diagnostic commands

```
ceph osd pool ls detail
ceph osd crush rule dump
ceph osd map vm-rbd test-object
ceph osd map .mgr test-object
ceph pg stat
ceph pg dump pgs_brief
ceph tell osd.3 dump_ops_in_flight
ceph tell osd.5 dump_ops_in_flight
ceph tell osd.7 dump_ops_in_flight
```

Pool reset and new RBD pool

```
for node in prox22 prox23 prox24; do
  ssh root@"$node" 'systemctl stop pvestatd'
done
ceph config set mon mon_allow_pool_delete true
ceph osd pool delete vm-rbd vm-rbd --yes-i-really-really-mean-it
ceph osd pool delete .mgr .mgr --yes-i-really-really-mean-it

# .mgr was recreated automatically
ceph osd pool create ceph 32 32 replicated replicated_rule
ceph osd pool set ceph size 3
ceph osd pool set ceph min_size 2
ceph osd pool application enable ceph rbd
rbd pool init ceph

ceph config set mon mon_allow_pool_delete false
```

Result: 33 PGs active+clean and Ceph HEALTH_OK.

8. Proxmox Ceph storage integration

```
pvesm add rbd ceph --pool ceph --content images,rootdir
sed -n '/^rbd: ceph$/,/^[^[:space:]]/p' /etc/pve/storage.cfg
pvesm status
```

Final entry: storage ID ceph, RBD pool ceph, content rootdir and images.

9. PERC/JBOD validation for ZFS

All three Dell servers use a PERC H755 Adapter. The controller remains in RAID personality for the OS SSD RAID1, while all twelve 20 TB SAS HDDs are individually exposed as JBOD. No single-disk RAID0 virtual drives were present.

```
PERC=/usr/local/sbin/perccli64
"$PERC" show
"$PERC" /c0 show
"$PERC" /c0 /eall /sall show
"$PERC" /c0 /vall show
```

PERC item	Verified state
Controller	PERC H755 Adapter
Personality	RAID-Mode
OS media	2 x 893.750 GB SSD in RAID1, OS_Disk
Large HDDs	12 x 18.190 TB SAS HDD
Large HDD type	JBOD, DG = -
Controller health	Optimal

10. ZFS disk prechecks

```
for node in prox22 prox23 prox24; do
  ssh root@"$node" '
    lsblk -d -o NAME,SIZE,MODEL,SERIAL,WWN,PHY-SEC,LOG-SEC /dev/sd{e..k}
    for disk in /dev/sd{e..k}; do wipefs --no-act "$disk"; done
    ceph-volume lvm list | grep -E "/dev/sd[e-k]" || echo "No Ceph ownership"
    pvs --noheadings -o pv_name | grep -E "/dev/sd[e-k]" || echo "No LVM ownership"
    zpool status 2>/dev/null || true
    zpool import 2>/dev/null || true
  '
done
```

All candidate disks were clean, not owned by Ceph or LVM, individually identifiable by stable WWN paths, and reported 4096-byte physical sectors with 512-byte logical sectors. ashift=12 was selected.

11. RAIDZ2 creation on all three nodes

One independent seven-disk RAIDZ2 pool was created on each node using stable /dev/disk/by-id/wwn-* paths. Each pool uses double parity. /dev/sdl was kept outside the vdev as reserved capacity.

Creation pattern used on each node

```
ZFS_BY_ID=(
  /dev/disk/by-id/wwn-<disk-1>
  /dev/disk/by-id/wwn-<disk-2>
  /dev/disk/by-id/wwn-<disk-3>
  /dev/disk/by-id/wwn-<disk-4>
  /dev/disk/by-id/wwn-<disk-5>
  /dev/disk/by-id/wwn-<disk-6>
  /dev/disk/by-id/wwn-<disk-7>
)

zpool create -o ashift=12   -O compression=lz4 -O atime=off -O xattr=sa   -O acltype=posixacl -O mountpoint=none   loc

zfs create -o compression=lz4 -o atime=off -o xattr=sa   -o acltype=posixacl -o mountpoint=none local-zfs-<node>/pve
```

Proxmox local storage entries

```
pvesm add zfspool local-zfs-prox22 --pool local-zfs-prox22/pve   --content images,rootdir --nodes prox22 --sparse 1
pvesm add zfspool local-zfs-prox23 --pool local-zfs-prox23/pve   --content images,rootdir --nodes prox23 --sparse 1
pvesm add zfspool local-zfs-prox24 --pool local-zfs-prox24/pve   --content images,rootdir --nodes prox24 --sparse 1
```

Final ZFS results

Node	Pool	Raw pool size	Available	State
prox22	local-zfs-prox22	127T	84.7T	ONLINE
prox23	local-zfs-prox23	127T	84.7T	ONLINE
prox24	local-zfs-prox24	127T	84.7T	ONLINE

All pools reported zero read, write and checksum errors and no known data errors. Properties were ashift=12, compression=lz4, atime=off, xattr=sa and acltype=posix.

12. Physical slot mapping

Linux devices were correlated to PERC enclosure 252 slots using serial numbers. PERC detailed WWNs can differ in the final hexadecimal nibble from Linux WWNs; serial number is therefore the primary correlation value.

prox22 - enclosure 252

Slot	Linux disk	Serial	Linux WWN	Role
0	/dev/sda	ZYDGFEW6	0x5000c500f588273b	Ceph osd.0
1	/dev/sdb	ZYDG4WXP	0x5000c500f58885ab	Ceph osd.3
2	/dev/sdc	ZYDG0CL3	0x5000c500f5880ffb	Ceph osd.6
3	/dev/sdd	ZYDFZ35V	0x5000c500f58810f3	Ceph osd.9
4	/dev/sde	ZYDG62WJ	0x5000c500f5881f8b	ZFS RAIDZ2
5	/dev/sdf	ZYDG2VVR	0x5000c500f589219b	ZFS RAIDZ2
6	/dev/sdg	ZYDFV3PC	0x5000c500f587c8af	ZFS RAIDZ2
7	/dev/sdh	ZYDGDMT3	0x5000c500f5889713	ZFS RAIDZ2
8	/dev/sdi	ZYDGBV46	0x5000c500f588891f	ZFS RAIDZ2
9	/dev/sdj	ZYDG6AWH	0x5000c500f5893a8b	ZFS RAIDZ2
10	/dev/sdk	ZYDFWL9E	0x5000c500f58801ff	ZFS RAIDZ2
11	/dev/sdl	ZYDFW4AP	0x5000c500f588810b	Reserved

prox23 - enclosure 252

Slot	Linux disk	Serial	Linux WWN	Role
0	/dev/sda	ZYDFR75F	0x5000c500f5886013	Ceph osd.1
1	/dev/sdb	ZYDG3L2Q	0x5000c500f5892617	Ceph osd.4
2	/dev/sdc	ZYDGCX8R	0x5000c500f5892043	Ceph osd.7
3	/dev/sdd	ZYDGD4MM	0x5000c500f5878807	Ceph osd.10
4	/dev/sde	ZYDG1DK6	0x5000c500f588cdcf	ZFS RAIDZ2
5	/dev/sdf	ZYDG05BG	0x5000c500f588b7af	ZFS RAIDZ2
6	/dev/sdg	ZYDG2WPB	0x5000c500f58832bb	ZFS RAIDZ2
7	/dev/sdh	ZYDFYB0P	0x5000c500f588cf43	ZFS RAIDZ2
8	/dev/sdi	ZYDG252H	0x5000c500f588010b	ZFS RAIDZ2
9	/dev/sdj	ZYDGADQ1	0x5000c500f587afcf	ZFS RAIDZ2
10	/dev/sdk	ZYDDT9JV	0x5000c500f55169b3	ZFS RAIDZ2
11	/dev/sdl	ZYDG7ZYQ	0x5000c500f5892757	Reserved

prox24 - enclosure 252

Slot	Linux disk	Serial	Linux WWN	Role
0	/dev/sda	ZYDFWT0C	0x5000c500f587b6ab	Ceph osd.2
1	/dev/sdb	ZYDFMNJ2	0x5000c500f588928f	Ceph osd.5
2	/dev/sdc	ZYDEHZW4	0x5000c500f587aef7	Ceph osd.8
3	/dev/sdd	ZYDEML2L	0x5000c500f57444af	Ceph osd.11
4	/dev/sde	ZYDG9GLB	0x5000c500f5882a53	ZFS RAIDZ2
5	/dev/sdf	ZYDGDGGH	0x5000c500f587fe6b	ZFS RAIDZ2
6	/dev/sdg	ZYDB04XZ	0x5000c500f56ad933	ZFS RAIDZ2
7	/dev/sdh	ZYDGH3XB	0x5000c500f5880d57	ZFS RAIDZ2
8	/dev/sdi	ZYDFXRKY	0x5000c500f587ab8f	ZFS RAIDZ2
9	/dev/sdj	ZYDFT406	0x5000c500f587cc7b	ZFS RAIDZ2
10	/dev/sdk	ZYDG9GR2	0x5000c500f5880f23	ZFS RAIDZ2
11	/dev/sdl	ZYDFHHAE	0x5000c500f5887ebb	Reserved

13. Physical locate and tagging workflow

```
PERC=/usr/local/sbin/perccli64
# Example: enclosure 252, slot 4
"$PERC" /c0/e252/s4 start locate
# Apply and verify label
"$PERC" /c0/e252/s4 stop locate
```

The locate LED function was successfully tested. The combined locator script maps disk or partition input to the parent physical disk, detects Ceph/ZFS/reserved role, identifies the PERC slot by serial, proposes the tag, and controls the locate LED.

Combined script usage

```
chmod 750 /root/ceph-zfs-disk-locator.sh
/root/ceph-zfs-disk-locator.sh /dev/sda   # Ceph
/root/ceph-zfs-disk-locator.sh /dev/sde   # ZFS
/root/ceph-zfs-disk-locator.sh /dev/sdl   # Reserved
/root/ceph-zfs-disk-locator.sh /dev/sde1  # Partition -> parent disk
```

The script intentionally refuses unsafe physical location of the OS virtual disk /dev/sdm. The OS RAID1 members can be identified individually only through enclosure 251 slots 0 and 1.

Recommended physical tags

Format: **NODE-ROLE-ID-SLOT**, with serial number on the second line. Examples:

```
PROX24-CEPH-OSD2-S0
SN: ZYDFWT0C

PROX24-ZFS-R2-01-S4
SN: ZYDG9GLB

PROX24-RESERVED-S11
SN: ZYDFHHAE
```

14. Final validation commands

```
# Ceph
ceph -s
ceph health detail
ceph mon stat
ceph mgr stat
ceph osd stat
ceph osd tree
ceph pg stat
ceph osd pool ls detail

# ZFS and Proxmox storage
for node in prox22 prox23 prox24; do
  ssh root@"$node" 'zpool status; zpool list; zfs list; pvesm status'
done

# Safety
ceph config get mon mon_allow_pool_delete
```

15. Final operational status

Check	Result
Proxmox cluster quorum	PASS
Ceph monitor quorum	PASS
Ceph health	HEALTH_OK
Ceph OSDs	12 up / 12 in
Ceph PGs	33 active+clean
RBD storage ceph	Active
ZFS prox22	ONLINE, no known data errors
ZFS prox23	ONLINE, no known data errors
ZFS prox24	ONLINE, no known data errors
PERC JBOD mapping	PASS
Physical slot map	Completed
Locate LED test	PASS
Proxmox node restrictions	Correct

Operational guidance: Use Ceph RBD for shared and HA VM/LXC disks. Use each local RAIDZ2 pool only for workloads local to its owning node, or with separately configured replication/backup. RAIDZ2 protects against up to two disk failures within its local vdev, but not against total node failure.